MILITARY SPECIFICATION

TUBE, CENTRIFUGE, ROUND STEM BOTTOM, GRADUATED, 100 ML

1. SCOPE

1.1 This specification covers one type of centrifuge tube, suitable for use in the determination of water and sediment in crude mineral oils.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

DD-G-541	Glass (Laboratory).
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Boxes, Set-Up, Paperboard.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM Designation C148	Tentative Methods of Polariscopic Examination of Glass Containers.
ASTM Designation D96	Standard Methods of Test for Water and Sediment in Crude Oils.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

3. REQUIREMENTS

3.1 Material. Centrifuge tube shall be transparent borosilicate glass conforming to DD-G-541 for type I, class A.

3.2 Construction. Centrifuge tube shall be constructed in accordance with the design, marking, and dimensions shown for pear-shaped centrifuge tubes in ASTM Designation D96, except that area for marking may be enameled and inner diameter of the top of the tube shall be 15 $\pm$ 3 millimeters.

3.3 Capacity. Capacity and partial capacities shall be within the tolerances specified in ASTM Designation D96.

3.4 Resistance to centrifugal force. Centrifuge tube shall not break or crack, when tested as specified in 4.4.1.

3.5 Strain. Ring strain shall be not greater than temper No. 1, and any longitudinal strain, if present, shall be faint and highly diffused, when tested as specified in 4.4.2.

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3.6 Marking. Marking shall be legible, and marking and area for marking shall be permanent when tested as specified in 4.4.3.

3.7 Identification marking. Each tube shall be marked with the manufacturer's name or registered trademark.

3.8 Workmanship. Workmanship shall be first class throughout. Tubes shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	2.5
Minor defects	II	4.0
For dimensional examination	S-1	2.5

4.2.2 For Tests. Sampling for tests shall be conducted in accordance with MIL-STD-105.

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Component (unit of product)	Characteristic	Requirement	Test Procedure	Inspection Level	AQL
Tube	Resistance to centrifugal force	3.4	4.4.1	S-2	*
Tube	Strain	3.5	4.4.2	S-2	*
Tube	Permanency of marking	3.6	4.4.3	S-1	*

* Acceptance number shall be zero.

4.3 Examination. The tubes shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects:

Categories and defects*

Major

101 Tube not free of cracks.
102 Tube not free of chips.
103 Tube not free of deformity.
104 Tube not free of bubbles.
105 Graduation markings not complete, correct, or legible.
106 Area for marking missing.

Minor

201 Tube not free of scratches.
202 Manufacturer's identification not complete, correct, or legible.

* Examination shall not be restricted to the classified possible defects listed above.

4.3.2 Dimensional examination. The tubes shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

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4.4.1 Resistance to centrifugal force. Tubes shall be filled to the 100-milliliter graduation with water, placed in a centrifuge, and spun for 15 minutes at the maximum relative centrifugal force indicated in the ASTM Designation D96.

4.4.2 Strain. Tubes shall be tested in accordance with ASTM Designation C148.

4.4.3 Permanency of marking. Gloss on enameled area for marking may fade. If area can accept a pencil mark after testing, area shall be considered as having passed test.

4.4.3.1 Resistance to alkali. The centrifuge tube shall be immersed completely in 1N sodium hydroxide at room temperature, covered, boiled for $\frac{1}{2}$ hour, and allowed to stand in the alkali for an additional 1 hour. The centrifuge tube shall then be removed from the alkali, throughly rinsed with water, and dried by rubbing vigorously with a cloth.

4.4.3.2 Resistance to acid.

4.4.3.2.1 Hydrochloric acid. The centrifuge tube shall be immersed completely in concentrated hydrochloric acid (specific gravity 1.18) at room temperature, covered, boiled for $\frac{1}{2}$ hour, and allowed to stand in the acid for an additional 1 hour. The centrifuge tube shall then be removed from the acid, thoroughly rinsed with water, and dried by rubbing vigorously with a cloth.

4.4.3.2.2 Sulfuric acid. The centrifuge tube shall be immersed completely in concentrated sulfuric acid (specific gravity 1.84) at room temperature and allowed to stand 1 hour. The centrifuge tube shall then be removed, thoroughly rinsed with water, and dried by rubbing vigorously with a cloth.

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5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One (1) tube, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table I shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table I Packaging Quantities

Unit package	Intermediate package	Exterior container
1 units	6 units	(See 5.2)

5.1.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Unit package shall be a carton of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676, except that commercial colors shall be acceptable. Carton shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of, or damage to, the contents. Closure shall be adequate to prevent spillage of contents under normal handling.

5.1.3.2 Intermediate package. Intermediate package shall be a carton of appropriate size constructed in accordance with PPP-B-676, except that commercial colors shall be acceptable, or PPP-B-636. Closure shall be adequate to prevent spillage of contents under normal handling.

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5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather-resistant. The gross weight of wood boxes shall not exceed 200 pounds. The gross weight of fiberboard boxes shall not exceed the weight limitation of the box specification.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specifiied in the appendix, and waterproofing shall conform to 30.4, of PPP-B-636.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall conform to PPP-B-601, table I, using type III, class I plywood; or shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type CF, class domestic, and method I closure shall be utilized. The gross weight of wood boxes shall not exceed 200 pounds. The gross weight of fiberboard boxes shall not exceed the weight limitation of the box specification.

5.2.3 Level C. The commodity shall be packed in substantial commerical containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

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5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Marking shall include the legend:

"GLASS-HANDLE WITH CARE."

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review activities:
Army - MD
Navy - MS
Air Force - 03

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